Weekly Progress Update

Exploitation Benchmarks, Plot Refinement, and Game 1 Multi-Agent Runs

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What This Week Focused On

Primary effort: writing + analysis refinement + Game 1 multi-agent execution

Paper-writing updates

- **Related work:** expanded negotiation literature framing (strategy signatures, adaptive exploitability, test-time strategic compute, and multi-party benchmarks).
- **Approach:** tightened formalism for fairness benchmarks and added explicit $n > 2$ Game 1 design details (Proposal 1 invasion + Proposal 2 matched ecology).
- **Analysis:** expanded mechanism-level interpretation (especially Game 1 qualitative failure modes and cross-game interpretation structure).
- **Conclusion:** added the preliminary $n = 3$ signal (Schelling-point dynamic) with explicit caveat on current statistical power.

Experiment/plot updates

- Produced paired adversary-vs-baseline exploitation plots across all three games.
- Refined Game 1 $n > 2$ plotting and qualitative breakdown visualizations by adversary Elo.

Intuition: What "Fair Share" Means Here

Question: Given an agreed outcome, did a model get *more* or *less* than a principled fair benchmark?

Games 1 & 2 (item split / treaty)

- Fair benchmark: **Nash Bargaining Solution (NBS)**.
- Intuition: among feasible agreements, NBS is the point that balances bargaining power by maximizing the product of utilities.

Game 3 (co-funding)

- Fair benchmark: **Lindahl equilibrium** cost shares.
- Intuition: for each funded project, each agent pays in proportion to how much they value it.

Exploitation index

- Positive: extracted above fair share.
- Zero: at fair share.
- Negative: below fair share.

Technical Definitions (Equations)

NBS benchmark (Games 1 and 2)

$$\mathbf{A}^{\mathrm{NBS}} = \arg\max_{\mathbf{A}} \prod_{i=1}^{n} u_i(\mathbf{A}), \quad E_i^{\mathrm{NBS}} = \frac{u_i^{\mathrm{actual}} - u_i^{\mathrm{NBS}}}{u_i^{\mathrm{NBS}}}$$

Lindahl benchmark (Game 3)

$$x_{ij}^{\mathrm{L}} = c_j \cdot \frac{v_{ij}}{\sum_k v_{kj}}, \quad u_i^{\mathrm{L}} = \sum_{j \in S^*} \left(v_{ij} - x_{ij}^{\mathrm{L}} \right), \quad E_i^{\mathrm{L}} = \frac{u_i^{\mathrm{actual}} - u_i^{\mathrm{L}}}{|u_i^{\mathrm{L}}|}$$

In this deck

- **Adversary EI plots:** $i =$ adversary model.
- **Baseline EI plots:** $i =$ GPT-5-nano.
- X-axis is adversary Elo in both cases.

Qualitative Mechanism Map (11 Mechanisms)

How Elo translates (or fails to translate) into payoff

Game 1 (Item Allocation)

- 1) Protocol collapse at the floor (hallucinated partner / parser fallback)
- 2) Self-contradictory legal proposal at the floor (preference-incoherent splits)
- 3) Fairness-frame exploitation (lower-mid tier)
- 4) Symmetric anchoring with delay tax (upper-mid tier)
- 5) Anchor-and-drift with stable top-item capture (high tier)
- 6) $n = 3$ Schelling-point reframing (focal agent sets coordination focal point)

Game 2 (Diplomatic Treaty)

- 7) Concession-narration ratchet (self-recorded concessions become sticky)

Game 3 (Participatory Budgeting)

- 8) Structural deadlock under orthogonality + scarcity
- 9) Cognitive floor: pledging to zero-value projects
- 10) Noise-as-free-riding via unfunded-project pledges
- 11) High-tier coordination control (precise repair and burden asymmetry)

Quote Evidence: Game 1 Mechanisms

Protocol collapse at floor

> *. . . we can establish a clear path to Stone. . .*
> **Agent_2:** *I appreciate your willingness to adapt. . .*

Single Agent_1 message includes hallucinated Agent_2 continuation.

Fairness-frame exploitation (lower-mid)

> *For me, the Apple is the most important item. . . The Quill and Pencil are also items of interest. . . The Jewel and Stone are lower on my priority list.*

Jewel was actually above Pencil in that run's preference vector.

$n = 3$ **Schelling-point reframing**

> *The core issue: All three of us want Stone as our top priority. That means whoever gets Stone is getting a massive share of value, so the remaining items need to be distributed fairly to the other two.*

Quote Evidence: Game 2/3 Mechanisms

Game 2: concession-narration ratchet

> *I've moved fentanyl from 78% → 50% → 40%... You've moved fentanyl from 6% → 6% → 6%. Zero movement.*

Narrated concessions become hard to retract in subsequent rounds.

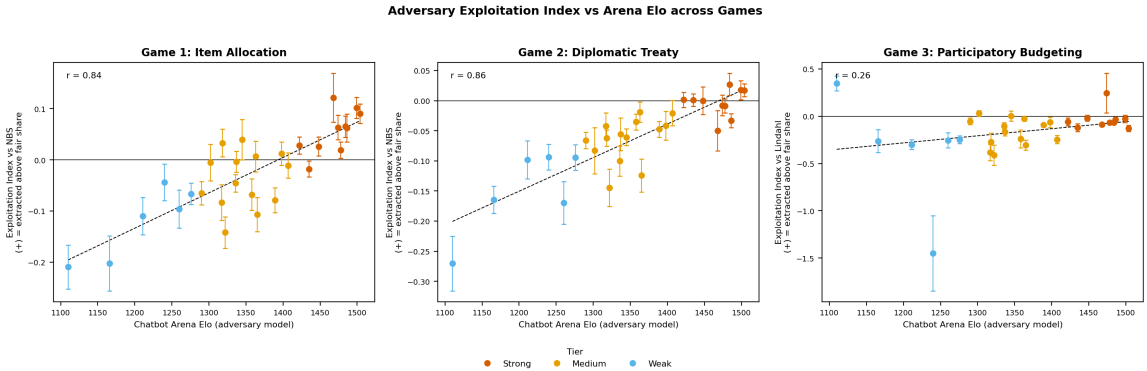
Game 3: structural deadlock / low-σ reasoning

> *Parkside (14 cost) is technically fundable this round, but I assign it* **zero value***, so prioritizing it would waste my budget.*

Game 3: high-tier coordination repair

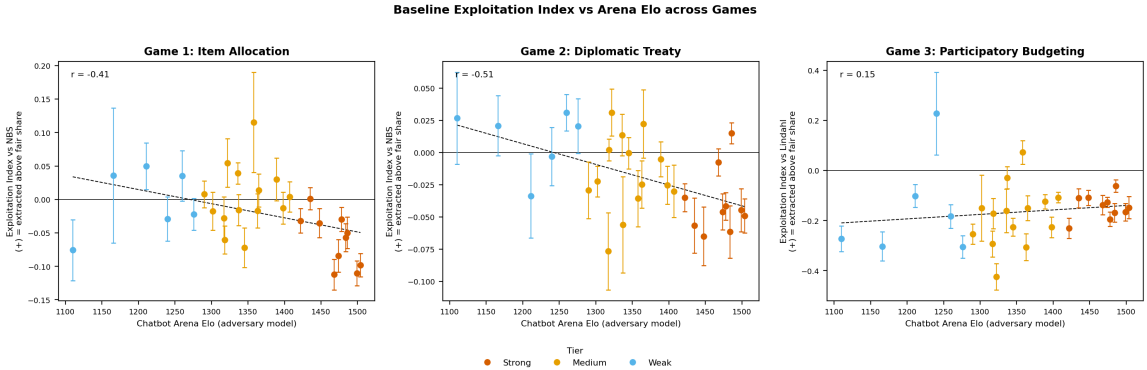
> *I propose we make our contributions precise. For example: I will contribute exactly 15 to the Market Street Bike Lane. You contribute exactly 21 to the Harborview Bus Shelter Canopies.*

Adversary Exploitation vs Elo (All Games)



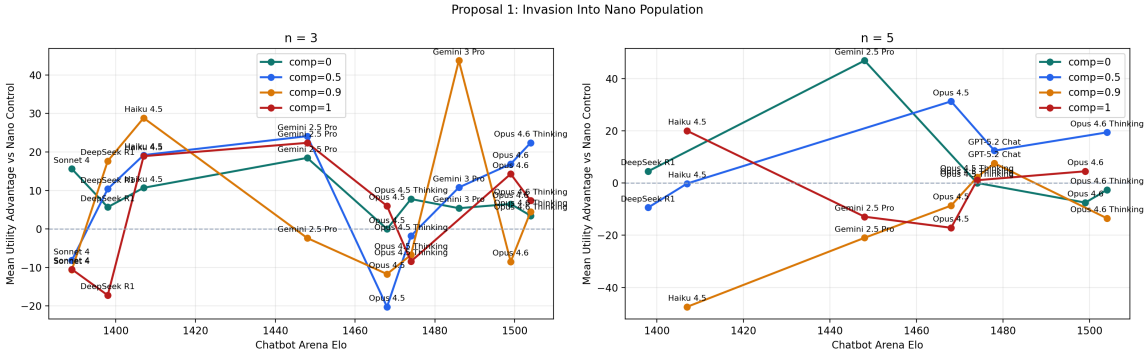
Readout: stronger adversaries extract more above fairness in Games 1 and 2 (high positive r), while Game 3 is weaker/noisier due to public-good coordination effects.

Baseline Exploitation vs Adversary Elo (All Games)



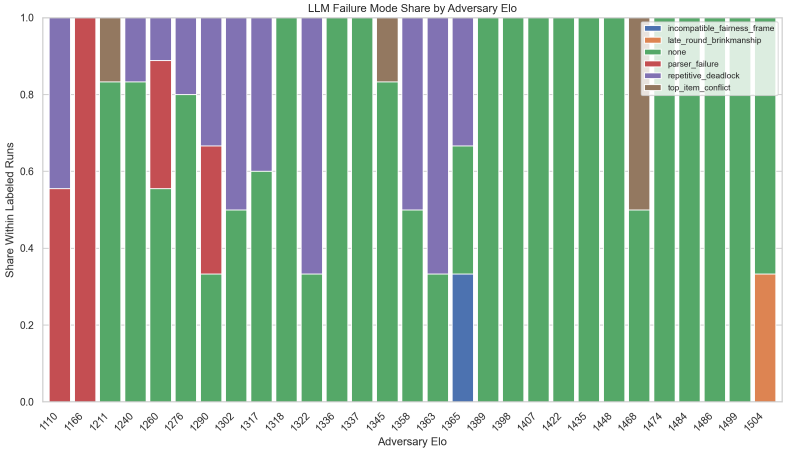
Readout: in Games 1 and 2, baseline fair-share deviation trends downward as adversary Elo rises; Game 3 remains mixed and closer to coordination-regime-dependent behavior.

Game 1 Multi-Agent Proposal 1 (Invasion Design)



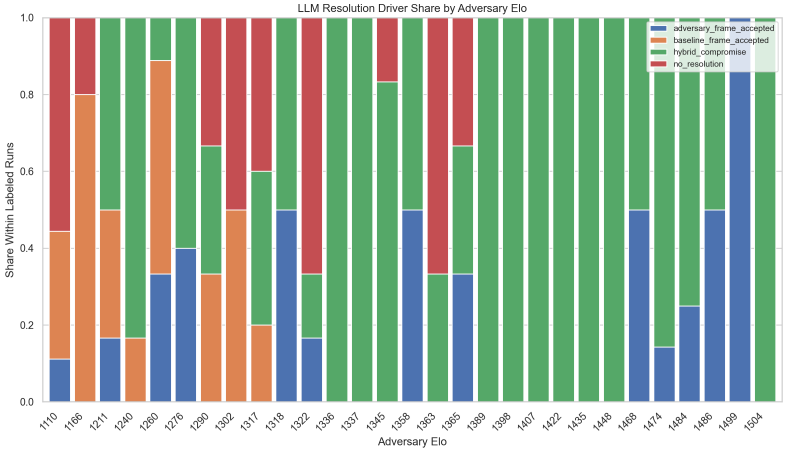
Readout: focal-model utility advantage versus nano control is condition-sensitive across competition levels; preliminary $n = 5$ behavior suggests coalition/coordination effects rather than simple dilution of capability.

Game 1 Qualitative Labels: Failure Modes by Elo



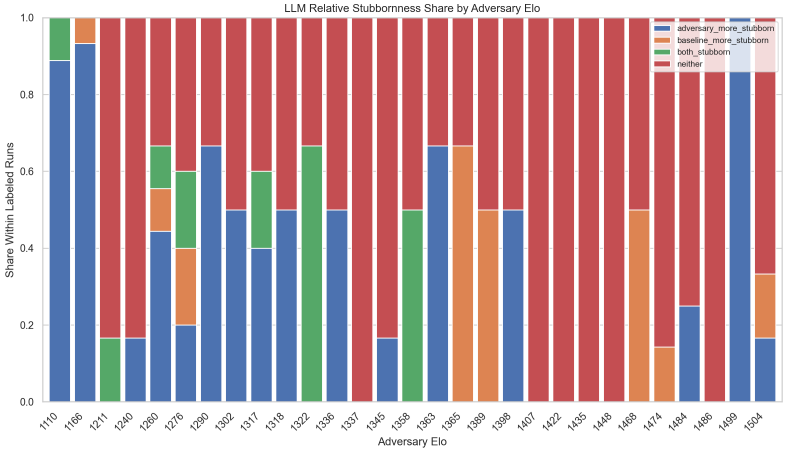
Readout: lower Elo bands show higher shares of parser/repetitive-deadlock failures; higher Elo runs are increasingly dominated by "none" (clean execution) with narrower failure diversity.

Game 1 Qualitative Labels: Resolution Drivers by Elo



Readout: hybrid compromise dominates in stronger bands; weaker bands show larger no-resolution and baseline-frame acceptance shares.

Game 1 Qualitative Labels: Relative Stubbornness by Elo



Readout: high-Elo interactions are often "neither stubborn" or asymmetrically adversary-stubborn depending on regime; low-tier runs contain more unstable stubbornness patterns.

Qualitative Label Audit: Haiku Coverage and Scope

Was this all transcripts? No. This was a stratified + extreme-case labeled subset.

- Full Game 1 run pool: **840** runs (‘run_count‘).
- Haiku-labeled subset: **127** runs (‘labeled_count‘, 15.1%).
- Judge model: `claude-haiku-4-5-20251001`.
- Label families per run: opening/adaptation style, failure mode, resolution driver, relative stubbornness.
- Sampling: `per-stratum=3` by (adversary tier, competition), plus `extreme-count=18` on high/low utility gap, high stubbornness, late rounds, and no-consensus high-stubbornness cases.

Counts used in slides 11–13 (n=127 labeled runs)

- **Failure mode:** none 78, parser_failure 24, repetitive_deadlock 19, top_item_conflict 3, late_round_brinkmanship 2, incompatible_fairness_frame 1.
- **Resolution driver:** hybrid_compromise 65, baseline_frame_accepted 26, no_resolution 20, adversary_frame_accepted 16.
- **Relative stubbornness:** neither 65, adversary_more_stubborn 43, both_stubborn 10, baseline_more_stubborn 9.

Slide 11 Audit: Failure-Mode Quote Exemplars

Concrete transcript examples used for each failure-mode label

`none`: "I agree that Option B is the best starting point for you, and I'm prepared to treat it as the leading framework."
(gpt-5.4-high, Elo 1484)

`parser_failure`: "Great opening, Agent_1. I share your emphasis on Quill and Stone..."
paired with 1.0 adversary parse-failure rate / proposer-gets-all fallback

`repetitive_deadlock`: "From rounds 1 through 9 we've learned a durable pattern... Pencil is the hinge..."
(deepseek-v3, repeated near-identical proposals)

`top_item_conflict`: "Pencil is the top priority for both of us... I'm not willing to concede Quill."
(claude-opus-4-5)

`late_round_brinkmanship`: "I'm not going to give Stone away without a very solid payoff."
(claude-opus-4-6-thinking)

`incompatible_fairness_frame`: "your continued framing... is exhausting and demonstrably unproductive... despite my clear and repeated statements..."
(gemma-3-27b-it impasse)

Transcript links and full snippets are in `analysis/game1_qualitative_review/game1_label_exemplar_index.md`.

Slide 12 Audit: Resolution-Driver Quote Exemplars

Concrete transcript examples used for each resolution-driver label

- `adversary_frame_accepted`: "Yes, I would vote yes to Apple + Pencil + Stone for Agent_1 and Quill + Jewel for me..."
  Baseline explicitly accepts the adversary package frame (gpt-5.4-high, Elo 1484).
- `baseline_frame_accepted`: "Thank you, Agent_1... reaffirming Option C as the proposed path forward."
  Adversary tracks and reaffirms baseline's option framing.
- `hybrid_compromise`: "I value Apple, Stone, and Pencil the most. Jewel and Quill have zero value to me, so I'm happy to trade them away."
  Complementary-preference package where both sides' priorities are integrated (deepseek-r1, Elo 1398).
- `no_resolution`: "I am formally declaring an impasse..."
  No shared package emerges by final round (gemma-3-27b-it).

Full context and transcript links: `analysis/game1_qualitative_review/game1_label_exemplar_index.md`.

Slide 13 Audit: Relative-Stubbornness Quote Exemplars

Concrete transcript examples used for each relative-stubbornness label

- `adversary_more_stubborn`: "last round, the votes showed you did not accept Option B..."
  Baseline adapts toward adversary package while adversary keeps anchor (gpt-5.4-high).
- `baseline_more_stubborn`: "your continued framing... is exhausting and demonstrably unproductive..."
  Adversary reports baseline repeatedly reasserting same frame (gemma-3-27b-it).
- `both_stubborn`: "From rounds 1 through 9... Pencil is the hinge..."
  Both sides keep returning to fixed anchor structures (deepseek-v3).
- `neither`: "My top priorities are Jewel and Quill..." + "I value Apple, Stone, and Pencil the most... happy to trade [the rest]."
  Low-friction complementary disclosure with quick consensus (deepseek-r1).

Full context and transcript links: `analysis/game1_qualitative_review/game1_label_exemplar_index.md`.

Synthesis

What these new results add

- We now have **both sides** of fairness deviation: adversary and baseline exploitation indices under the same benchmarks.
- Game 1/2 look like capability-linked extraction gradients; Game 3 remains more structure-driven (threshold public-good coordination).
- Preliminary $n > 2$ Game 1 results are compatible with a **Schelling-point coordination story** where weaker coalitions do not reliably block a stronger focal.

Immediate next steps

- Finish powering the ongoing $n > 2$ Game 1 batch and update confidence intervals.
- Add matched baseline/adversary EI tables per competition cell for appendix-ready reporting.
- Integrate these figures and definitions directly into the current analysis/conclusion narrative draft.